

About Certis

We are a leading security services organisation. We put technology to work in making the world a safer place. Our mission is that as trusted partners, we protect lives and assets, and deliver integrated critical services. By empowering our people, our goal is to enable our communities to stay safe and thriving. We offer a broad range of opportunities and career pathways for our people to grow and explore their potential. We believe that people, equipped with innovative thinking and technological capabilities will make the world safer, smarter and better.

关于 Certis

我们是一家领先的安保服务机构，致力于运用科技让世界变得更安全。我们的使命是：作为值得信赖的合作伙伴，我们保护生命与资产，并提供整合性的关键服务。通过赋能员工，我们希望帮助社区保持安全并持续繁荣。

我们为员工提供多元的发展机会和职业路径，支持他们探索并发挥自身潜力。我们深信，具备创新思维与技术能力的人才，将能够让世界变得更安全、更智能、更美好。

Note:

Full time job opportunities (for graduates) and internship (for undergraduates) available with multiple vacancies in Singapore & Hangzhou. JD below.

新加坡与杭州均提供全职岗位（面向毕业生）及实习岗位（面向本科生），且设有多个名额。职位说明如下。

1. Robotics Software (Fleet / Platform / Autonomy Integration)
2. Robotics AI (Perception / Learning / World Models)
3. Advanced Research (Robotics Foundation Models & Control)
4. Robotics and Embodied AI Strategic Partnership
5. AI Agent Development (Agent Orchestration & Long-term Memory)
6. Computer Vision (Spatial Intelligence & VLM Optimization)

Role 1: Robotics Software (Fleet / Platform / Autonomy Integration)

Overview

We are building physical intelligence for public and shared spaces (airports, hospitals, malls, campuses) using real robot fleets. This internship focuses on the core robotics software that makes heterogeneous robots operate safely and reliably at scale: fleet mission execution, robot integration, autonomy troubleshooting, and operational tooling.

What you will work on (you may choose 1–2 tracks listed below)

Track A: Robot Integration & Mission Runtime

- Implement robot adapters (command/control, health, telemetry) for wheeled robots, quadrupeds, or humanoids
- Build mission execution logic, error recovery, and teleoperation hooks

Track B: Fleet Platform Backend & Ops Tooling

- Build service APIs, mission/event schemas, and telemetry pipelines
- Create dashboards, incident replay tools, and integration tests for fleet reliability

Track C: Autonomy Integration & Field Debug (Navigation/Localization/Mapping)

- Build “autonomy triage” tooling: log parsers, health metrics, failure classifiers
- Create site acceptance scripts and reproducible test scenarios for navigation robustness

Track D: QA Automation

- Build regression harnesses, scenario replays, and automated smoke tests for robot releases

Expected deliverables

- A working software component that ships: e.g., a robot adapter, a telemetry dashboard, an autonomy debug toolkit, or a regression test suite. Technical report.

Requirements

- Strong programming in Python and/or C++ (other backend languages welcome)
- Comfort debugging complex systems (logs, networking, distributed components)
- Solid fundamentals in algorithms and software engineering
- Bonus: experience with ROS/ROS2, Linux, Docker, or robotics coursework/projects

What we value

- Ability to turn ambiguous problems into shipped, testable software
- Strong engineering discipline: clear code, reproducible tests, good documentation
- Curiosity about real-world robotics constraints (safety, reliability, latency)

How to apply

Please send:

- Resume + GitHub/portfolio (if available)

- 1–2 paragraphs: your preferred track(s) and a past project most relevant to this role

机器人软件（车队/平台/导航集成）

岗位概述

我们正在面向**公共/共享空间**（机场、医院、商场、园区等）打造可规模化部署的“具身智能”系统，使用真实的机器人车队在现场长期运行。本实习岗位聚焦于**核心机器人软件工程**：车队任务执行与管理、不同机器人平台的接入与适配、导航/定位/建图问题的排查与集成，以及运营所需的工具链。

方向（可选择 1–2 个方向）

方向 A：机器人接入与任务运行时（Adapter & Mission Runtime）

- 为轮式机器人/机器狗/人形机器人实现控制与状态适配（指令、健康状态、电量、遥操作接口、遥测数据）
- 实现任务执行逻辑、错误恢复机制、以及异常情况下的安全处理

方向 B：车队平台后端与运营工具（Backend & Ops Tooling）

- 开发服务端 API、任务/事件数据结构、遥测数据管道
- 构建运行监控看板、事件回放/复盘工具、集成测试与回归测试

方向 C：导航集成与现场排障（Navigation/Localization/Mapping Debug）

- 开发“导航问题排查工具箱”：日志解析、健康指标、故障分类
- 编写站点验收脚本与可复现测试场景，用于提升导航鲁棒性

方向 D：QA 自动化

- 搭建回归测试框架、场景回放、自动化冒烟测试，保障机器人版本质量

预期产出

- 可落地、可集成的组件：例如机器人的适配、遥测/监控模块、导航排障工具、回归测试套件等。技术报告。

基本要求

- 扎实的 **Python** 和/或 **C++** 编程能力（有后端语言经验欢迎）
- 擅长系统调试（日志分析、网络问题、分布式组件定位）
- 良好的软件工程基础与代码质量意识
- 加分项：**Linux**、**Docker**、**ROS/ROS2**、机器人相关课程或项目经验

看重的能力

- 把问题拆解成清晰目标并交付可用软件
- 工程纪律强：代码可读、测试可复现、文档清晰
- 对真实场景机器人限制（安全、可靠性、延迟、现场复杂性）有兴趣

申请方式

请提供：

- 简历 + GitHub/作品集（如有）
- 1-2 段简短说明：你最想做的方向（可多选）以及你最相关的一个项目/经历

Role 2: Robotics AI (Perception / Learning / World Models)

Overview

We are building physical intelligence for public and shared spaces (airports, hospitals, malls, campuses) using real robot fleets. This internship focuses on the AI and learning stack behind perception, skill learning, simulation/digital twins, and the data flywheel that turns real-world operations into continuously improving models.

What you will work on (you may choose 1–2 tracks listed below)

Track A: Perception Evaluation (Detection/Tracking Metrics + Dataset)

- Curate/clean datasets from robot and sensor data; define labeling guidelines
- Build evaluation pipelines for detection/tracking (e.g., precision/recall, IDF1/MOTA), analyze failure modes and false alarms
- Produce actionable reports to improve models and deployment quality

Track B: Manipulation Learning Assistant (BC / VLA Fine-tuning Support)

- Prepare training data (demonstrations, trajectories), run training experiments (behavior cloning, fine-tuning)
- Build task-level evaluation harnesses (success rate, time-to-complete, safety constraints)
- Help iterate on sim-to-real experiments and document learnings

Track C: Data/MLOps (Pipeline + Registry + Evaluation Runner)

- Implement dataset versioning, experiment tracking, and reproducible training/evaluation runs
- Prototype a lightweight model registry and automated evaluation runner
- Support packaging and deployment workflows for edge inference (robots/edge compute)

Track D: Simulation / Digital Twin

- Build simulation assets and benchmarks for robotics tasks and public-space scenarios
- Create domain randomization tools and scenario libraries for robust training/testing
- Integrate sim evaluation results with real-world logs to close the loop

Expected deliverables

- A working AI tooling or model-development component that can be integrated into the team's pipeline
- Technical report

Requirements

- Strong Python skills; experience with PyTorch
- Solid ML fundamentals (training, evaluation, debugging)
- Comfortable working with data pipelines and experimental rigor
- Bonus: CV/tracking experience, RL/imitation learning, simulation tools (Isaac Sim/MuJoCo/Gazebo), Docker

What we value

- Ability to turn research ideas into measurable, repeatable engineering outcomes
- Strong experimental discipline: ablations, metrics, reproducibility
- Curiosity about real-world robotics constraints (noise, lighting, occlusion, latency, safety)

How to apply

Please send:

- Resume + GitHub/portfolio (if available)
- 1-2 paragraphs: your preferred track(s) and one relevant project (paper/project link welcomed)

机器人 AI（感知 / 学习 / 世界模型）

岗位概述

我们正在打造可规模化部署的具身智能系统，使用真实机器人长期运行。本实习岗位聚焦机器人系统的 **AI 与学习能力**：数据集、操作技能学习（模仿学习/策略微调/强化学习）、数据与训练工程（Data/MLOps）、以及仿真/数字孪生，推动“数据 → 训练 → 评测 → 部署”的闭环迭代。

方向选择（可选 1-2 个方向）

方向 A：数据集

- 数据集整理/清洗与标注规范制定
- 搭建 AI 模型的评测流程（例如 precision/recall、IDF1/MOTA），分析失败模式
- 输出可执行的评测报告，改进建议，支撑上线模型的质量

方向 B：操作技能学习（BC / VLA 微调支持）

- 准备训练数据（示教轨迹/操作数据），运行训练实验（BC、Finetuning 等）
- 建立评测框架（成功率、完成时间、安全约束与失败类型）
- 参与 sim-to-real 迭代

方向 C：数据与 MLOps（数据管道 + 模型 评测自动化）

- 实现数据集版本管理、可追踪可复现的训练/评测流程。
- 模型管理（model registry）与自动评测。
- 边缘推理部署流程（机器人/Edge 的模型打包、评测与回滚）

方向 D：仿真 / 数字孪生

- 构建任务与场景的仿真资产，场景库，用于鲁棒训练/测试
- 将仿真评测与真实日志对齐，形成训练与回归验证闭环

预期产出

- 交付一个可集成的 AI 工具链或模型开发组件
- 技术报告：问题、方法、结果、下一步计划

基本要求

- 扎实的 **Python** 编程能力；具备 **PyTorch** 使用经验
- 良好的机器学习基础（训练、评测、调参、排错）
- 有数据处理与实验严谨性意识（指标、对比实验、可复现）
- 加分项：CV/跟踪经验、RL/模仿学习、仿真工具（Isaac Sim/MuJoCo/Gazebo）、Docker

我们看重的能力

- 能把研究/想法落到可量化、可复现、可交付的工程结果
- 实验纪律强：指标清晰、对比、复现实验流程
- 理解真实机器人场景限制（噪声、遮挡、光照变化、延迟、安全）

申请方式

请提供：

- 简历 + GitHub/作品集（如有）
- 1-2 段说明：你最想做的方向（可多选）以及一个相关项目（论文/项目链接欢迎）

Role 3: Advanced Research (Robotics Foundation Models & Control)

Overview

We are building physical intelligence for public and shared spaces (airports, hospitals, malls, campuses) using real robot fleets. This research internship focuses on forward-looking topics that can become the next generation of robotics capability: foundation models for perception and action, learning-based navigation, world models, and multi-robot coordination. You will work with our team to turn research into measurable prototypes and clear technical conclusions.

What you will work on (you may choose 1 track listed below)

Track A: Manipulation Research (VLA + RL)

- Explore one focus area: generalist manipulation, contact-rich robustness (slip/misalignment/occlusion recovery), or skill composition
- Train and evaluate policies using real data and/or simulation, with clear success metrics and ablations

Track B: Learning-based Navigation (VLN / End-to-End)

- Prototype learning-based navigation components (e.g., visual goal conditioning, traversability, re-localization aids)
- Assess feasibility for real deployments: reliability, safety, compute, and generalization

Track C: World Models (Self-supervised Video / Multimodal Predictive Models)

- Build predictive representation learning pipelines from video and multimodal streams
- Demonstrate measurable downstream benefit (forecasting, anomaly understanding, robustness)

Track D: Multi-robot Coordination (Task Allocation / Policies / Congestion)

- Develop algorithms for multi-robot scheduling and coordination under constraints
- Evaluate with simulators and log replays; define policy constraints and failure modes

Expected deliverables

- A working research prototype (code + reproducible experiments)
- A concise research memo: problem framing, method, experimental results, key takeaways, and next-step recommendations
- Optional (if results are strong): a draft workshop/paper outline or open-source-ready artifact

Requirements

- Strong research background evidenced by papers or high-quality projects (robotics/ML preferred)
- Excellent Python skills and experience with PyTorch/JAX
- Ability to design rigorous experiments (baselines, metrics, ablations, reproducibility)

- Comfort working with large-scale data and/or simulation
- Bonus: prior work in manipulation, RL/IL, navigation/VLN, video modeling, or multi-agent systems

What we value

- Choosing tractable problems that matter for real robots
- Experimental discipline and honest reporting of results
- Ability to bridge research and engineering (clean code, reproducible pipelines)

How to apply

Please send:

- Resume + Google Scholar / papers / GitHub (as applicable)
- A short statement: preferred track and what you want to prove/build
- (Optional) research summary of your best project/paper

高级研究（机器人基础模型与控制）

岗位概述

我们正在面向**公共/共享空间**（机场、医院、酒店、商场、园区等）打造可规模化部署的具身智能系统，使用真实机器人长期运行。本研究型实习聚焦更前沿、更具长期价值的方向：**机器人基础模型（感知与动作）、学习式导航、世界模型、多机器人协同**。你将与团队一起把研究想法落到**可复现的原型与可量化的结论**上。

研究方向（请选择 1 个主方向）

方向 A: Manipulation 研究（VLA + RL）

- 选择一个具体聚焦点：比如，通用操作、接触丰富场景的鲁棒性、技能组合、推理
- 基于真实数据和/或仿真训练策略

方向 B: 学习式导航（VLN / 端到端）

- 原型化学习式导航模块（例如视觉目标条件、可通行性评估、重定位辅助等）
- 评估落地可行性：可靠性、安全性、算力需求、跨场景泛化能力

方向 C: 世界模型（自监督视频 / 多模态预测模型）

- 构建基于视频与多模态流的预测表征学习流程
- 证明对机器人下游任务有可量化收益

方向 D: 多机器人协同（任务分配 / 策略约束 / 拥堵与资源调度）

- 研究基于 AI 和 Agent 的多机器人调度与协同算法，在约束条件下优化效率与安全
- 用仿真与评测，失败模式与可部署边界

预期产出

- 可运行的原型（代码 + 可复现实验）
- 研究文档：问题定义、方法、实验结果、关键结论、下一步建议

- （可选）若结果突出：论文，开源项目

基本要求

- 有扎实研究背景：论文或高质量项目
- 熟练 Python，并有 PyTorch/JAX 使用经验
- 具备严谨实验能力（基线对比、指标设计、消融实验、可复现性）
- 能处理大规模数据和/或仿真环境
- 加分项：操作、RL/模仿学习、导航/VLN、视频建模、多智能体系统相关经验

我们看重的能力

- 判断力：能选择对真实机器人有价值、且可在周期内推进的问题
- 能连接研究与工程：代码干净、流程可复用、结果可落地

申请方式

请提供：

- 简历 + Google Scholar / 论文链接 / GitHub（如有）
- 简短说明：你选择的方向，以及你想证明/构建的核心目标
- （可选）研究总结：你最好的一个论文或项目（方法与结果要点）

Role 4: Robotics and Embodied AI Strategic Partnership

Reporting To Department

Vice President, Head of Strategic Partnerships, Robotics

Internship Objective

To support strategic partnership development in robotics and embodied AI through market research, partnership analysis, and commercial engagement support. The intern will work closely with senior leadership to evaluate technology partners, identify use cases, and support discussions with global robotics and AI companies.

Key Responsibilities

Market and Industry Research

- Conduct structured research on global and China-based robotics and embodied AI companies, including humanoids, quadrupeds, autonomous mobile robots, and embodied intelligence platforms
- Track industry trends across technology maturity, commercialization progress, pricing, and deployment use cases
- Analyze competitive landscapes, including OEMs, system integrators, and AI software providers
- Prepare concise briefing materials, market maps, and comparison tables for senior management discussions
- Monitor ecosystem developments relevant to robotics and AI adoption in Asia and globally

Partnership and Strategy Support

- Support evaluation of potential strategic partners across hardware, AI software, and systems integration
- Assist in preparing partnership discussion materials including value propositions, collaboration models, and MOU summaries
- Support internal coordination across robotics, AI, engineering, and commercial teams to align partnership objectives
- Document meeting outcomes, key discussion points, and follow-up actions from partner engagements
- Conduct basic commercial and strategic analysis to support decision-making on exclusivity, distribution, and joint development models

Commercial Engagement Support

- Assist in developing use case narratives, pitch decks, and one-page summaries for customer engagements
- Conduct customer and sector research to identify high-potential application areas for robotics adoption
- Support pilot and proof-of-concept planning by documenting objectives, success criteria, and partner roles
- Support preparation for meetings with enterprise and government customers involving robotics and embodied AI solutions

- Help track engagement progress and feedback to inform future partnership and product strategy

Learning Outcomes

By the end of the internship, the intern will gain:

- First-hand exposure to real-world robotics and embodied AI strategic partnership at an enterprise scale
- Experience working directly with senior leadership on strategic partnerships and commercial decision-making
- Strong understanding of the global robotics ecosystem across hardware, AI, and deployment models
- Practical skills in market research, strategic analysis, and executive-level communication

Ideal Candidate Profile

- Undergraduate or postgraduate student from Zhejiang University, preferably in Robotics, Computer Science, Artificial Intelligence, Engineering, Economics, or Business
- Strong interest in robotics, embodied AI, and technology commercialization
- Excellent research, analytical, and synthesis skills
- Clear written and verbal communication skills in English; Mandarin proficiency is a strong advantage
- Highly organized, detail-oriented, and comfortable working with senior stakeholders
- Curious, proactive, and able to handle ambiguous, early-stage strategic topics

How to apply

Please send:

- Resume
- An essay in English 300 to 500 words to explain why you want to apply for this internship
- (Optional) research summary of your understanding of robotics and embodied AI ecosystem

职位名称: 机器人与具身智能战略合作

汇报部门

副总裁，机器人与具身智能战略合作负责人

实习目标

通过市场与行业研究、战略合作支持及商业拓展支持，协助推进机器人与具身智能领域的战略合作。实习生将直接与高级管理层合作，参与评估全球机器人与 AI 合作伙伴，梳理应用场景，并支持与产业伙伴的交流与合作。

主要职责

市场与行业研究

- 对全球及中国的机器人与具身智能企业开展系统性研究，涵盖人形机器人、四足机器人、自主移动机器人及具身智能平台
- 跟踪行业趋势，包括技术成熟度、商业化进展、价格水平及落地应用场
- 分析竞争格局，涵盖整机厂商、系统集成商及 AI 软件公司
- 为管理层会议准备简明的研究简报、市场地图及对比分析材料
- 持续关注与机器人及 AI 相关的政策、监管及产业生态变化

战略合作与伙伴支持

- 支持潜在战略合作伙伴的评估工作，覆盖硬件、AI 软件及系统集成方向
- 协助准备合作讨论材料，包括价值主张、合作模式及 MOU 摘要
- 协助内部协调机器人、AI、工程及商业团队，确保合作目标一致
- 记录合作方会议要点、关键结论及后续行动事项
- 开展基础商业及战略分析，支持关于独家合作、分销模式及联合开发的决策

商业拓展与客户支持

- 协助撰写应用场景说明、客户展示材料及一页式方案简介
- 针对重点行业与客户开展研究，识别高潜力机器人应用领域
- 支持试点项目与概念验证的规划，包括目标、成功标准及合作分工
- 支持与企业及政府客户的机器人与具身智能相关交流
- 汇总客户反馈与项目进展，为后续产品及合作策略提供输入

实习收获

实习结束后，实习生将获得：

- 直接参与企业级机器人与具身智能战略合作实践的第一手经验
- 与高级管理层合作，参与战略合作与商业决策过程
- 对全球机器人产业在硬件、AI 及落地模式方面的系统性理解
- 市场研究、战略分析及高层沟通能力的显著提升

理想候选人画像

- 浙江大学在读本科或研究生，专业包括但不限于机器人、计算机科学、人工智能、工程、经济或商科
- 对机器人与具身智能及其商业化应用具有浓厚兴趣
- 具备良好的研究能力、分析能力与信息整合能力
- 具备优秀的中英文书面与口头表达能力，普通话流利，英文能力为加分项
- 工作细致、有条理，能够与高级管理层高效协作
- 具备主动性与好奇心，能够适应探索性和早期阶段的战略议题

请提交：

- 个人简历

- 一篇 300 至 500 字的英文短文，阐述您申请此实习岗位的理由
- （可选）您对机器人和具身人工智能生态系统的理解的研究总结

Role 5: AI Agent Development (Agent Orchestration & Long-term Memory)

Overview

We are building intelligent AI agent systems that power Certis's next-generation security and operations products — including Mozart, SeVU, MIRA, VOXIS, and IncidentIQ. This internship focuses on building agents that can reason over complex tasks, use tools reliably, maintain persistent memory, and coordinate with other agents — grounded in real enterprise use cases such as incident management, security operations, and facility management.

The state of the art in AI agents is evolving rapidly — from prompt-chained workflows to systems with planning, reflection, and autonomous tool use. We are looking for interns who can work at this frontier and bring the latest techniques into production.

What you will work on (you may choose 1–2 tracks)

Track A: Agent Orchestration & Multi-Agent Systems

- Design and implement agentic workflows with planning, tool use, and self-correction loops for complex operational tasks
- Build multi-agent architectures with role specialization, delegation, and structured communication (e.g., supervisor-worker, debate, consensus patterns)
- Implement reliable tool integration: API calling, structured output parsing, error recovery, and retry logic
- Explore agentic design patterns: ReAct, plan-and-execute, reflection, tree-of-thought, and dynamic re-planning

Track B: Long-term Memory & Knowledge Architecture

- Design and implement memory systems that go beyond simple RAG: episodic memory (what happened), semantic memory (what the agent knows), and procedural memory (how to do things)
- Build retrieval pipelines that are context-aware, time-sensitive, and support memory consolidation and forgetting
- Explore emerging memory patterns: memory-augmented generation, self-reflective summarization, experience replay for agents
- Evaluate memory strategies on recall precision, task performance lift, and robustness over long interaction horizons

Track C: Agent Evaluation, Reliability & Safety

- Build evaluation frameworks that go beyond single-turn accuracy: multi-step task completion, tool-use correctness, reasoning faithfulness, and cost efficiency
- Design benchmark scenarios grounded in Certis operational workflows with realistic edge cases and failure modes
- Develop guardrails and safety mechanisms: output validation, hallucination detection, human-in-the-loop escalation, and graceful degradation
- Instrument agent systems for observability: tracing, token usage, latency profiling, and decision logging

Expected deliverables

- A working agent component or system that integrates into the team's product pipeline: e.g., a multi-agent orchestration module, a memory architecture, or an evaluation/observability harness
- Technical report: problem framing, architecture decisions, experiments, results, and production recommendations

Requirements

- Strong Python skills; hands-on experience building applications with LLM APIs (OpenAI, Anthropic, or open-source models)
- Familiarity with agent frameworks (LangChain, LangGraph, CrewAI, AutoGen, or similar) — but more importantly, understanding of the underlying patterns beyond any single framework
- Solid understanding of LLM capabilities and limitations: prompting strategies, function/tool calling, context window management, structured outputs
- Understanding of retrieval systems: vector databases, embedding models, chunking strategies, hybrid search
- Bonus: experience with agent observability tools (LangSmith, Arize, etc.), knowledge graphs, or deploying LLM-based systems in production

What we value

- **Systems thinking:** Agents are systems, not just prompts. We value people who think about failure modes, latency budgets, cost, and how components interact end-to-end
- **First-principles reasoning:** The ability to evaluate new techniques critically rather than chasing hype — understanding why something works, not just that it works
- **Engineering discipline:** Modular code, clear abstractions, reproducible experiments, and proper evaluation before claiming something works
- **Rapid prototyping with production awareness:** Move fast in experiments, but design with deployment constraints (latency, cost, reliability, safety) in mind
- **Curiosity about the frontier:** The agent landscape changes weekly — we value people who stay current with new papers, frameworks, and architectural patterns

How to apply

Please send:

- Resume + GitHub/portfolio (if available)
- 1–2 paragraphs: your preferred track(s) and a past project most relevant to this role. If you've built any agent system (even a personal project), tell us about it — what worked, what broke, and what you learned.

岗位五：AI Agent 开发（Agent 编排 与 长期记忆）

岗位概述

我们正在构建智能 AI Agent 系统，驱动 Certis 下一代安防与运营产品——包括 SeVU、Mozart、MIRA、VOXIS 和 IncidentIQ。本实习岗位聚焦于构建能够进行复杂任务推理、

可靠使用工具、维持持久记忆、并与其他 Agent 协作的系统——以事件管理、安防运营、设施管理等真实企业场景为基础。

AI Agent 领域正在快速演进——从简单的 prompt 链式 workflow，到具备规划、反思和自主工具使用能力的系统。我们寻找能够在这—前沿工作、并将最新技术落地到产品中的实习生。

方向选择（可选 1–2 个方向）

方向 A：Agent 编排与多 Agent 系统

- 设计并实现具备规划、工具调用和自我修正能力的 Agent workflow，应用于复杂运营任务
- 构建多 Agent 架构：角色分工、任务委派、结构化通信（如 supervisor-worker、辩论式、共识模式等）
- 实现可靠的工具集成：API 调用、结构化输出解析、错误恢复与重试逻辑
- 探索 Agent 设计模式：ReAct、plan-and-execute、反思、tree-of-thought、动态重规划等

方向 B：长期记忆与知识架构

- 设计并实现超越简单 RAG 的记忆系统：情景记忆（发生了什么）、语义记忆（Agent 知道什么）、程序记忆（如何做事）
- 构建上下文感知、时间敏感的检索管道，支持记忆整合与遗忘机制
- 探索前沿记忆模式：记忆增强生成、自反思摘要、Agent 经验回放
- 在召回精度、任务性能提升、长周期交互鲁棒性等维度评估记忆策略

方向 C：Agent 评测、可靠性与安全

- 构建超越单轮准确率的评测框架：多步骤任务完成、工具调用正确性、推理忠实度、成本效率
- 基于 Certis 运营场景设计基准测试，包含真实边缘场景与失败模式
- 开发护栏与安全机制：输出验证、幻觉检测、人机协同升级、优雅降级
- 构建 Agent 可观测性工具：链路追踪、Token 用量、延迟分析、决策日志

预期产出

- 可集成到团队产品管道的 Agent 组件或系统：例如多 Agent 编排模块、记忆架构、评测/可观测性工具
- 技术报告：问题定义、架构决策、实验结果、产品化建议

基本要求

- 扎实的 Python 编程能力；有使用 LLM API（OpenAI、Anthropic 或开源模型）构建应用的实践经验

- 熟悉 Agent 框架（LangChain、LangGraph、CrewAI、AutoGen 等）——更重要的是理解框架背后的设计模式，而非仅限于某一个框架
- 深入理解 LLM 的能力与局限：prompt 策略、function/tool calling、上下文窗口管理、结构化输出
- 理解检索系统：向量数据库、Embedding 模型、分块策略、混合检索
- 加分项：Agent 可观测性工具（LangSmith、Arize 等）、知识图谱、或 LLM 系统的生产环境部署经验

我们看重的能力

- **系统性思维**：Agent 是系统，不仅仅是 prompt。我们看重能够思考失败模式、延迟预算、成本以及组件端到端交互的人
- **第一性原理思维**：能够批判性地评估新技术，而非追逐热点——理解“为什么有效”而非仅仅“知道它有效”
- **工程纪律**：模块化代码、清晰抽象、可复现实验，在宣称有效之前先做充分评测
- **快速原型 + 产品化意识**：实验中快速迭代，但设计时考虑部署约束（延迟、成本、可靠性、安全）
- **对前沿的好奇心**：Agent 领域每周都在变化——我们看重持续跟踪最新论文、框架和架构模式的人

申请方式

请提供：

- 简历 + GitHub/作品集（如有）
- 1-2 段简短说明：你最想做的方向（可多选）以及你最相关的一个项目/经历。如果你构建过任何 Agent 系统（即使是个人项目），请告诉我们——什么有效、什么出了问题、你学到了什么。

Role 6: Computer Vision (Spatial Intelligence & VLM Optimization)

Overview

We are advancing the visual intelligence capabilities behind Certis's AI products — including SeVU (intelligent video analytics), Mozart, MIRA, VOXIS, and IncidentIQ. This internship focuses on spatial understanding, vision-language model (VLM) optimization, spatial-temporal behavior analysis, and scene intelligence for security and operations in public spaces. You will work on pushing the boundaries of what our vision systems can perceive, understand, and reason about.

What you will work on (you may choose 1–2 tracks)

Track A: Spatial Intelligence & Scene Understanding

- Develop or improve spatial reasoning capabilities: depth estimation, 3D scene understanding, spatial relationship extraction from camera feeds
- Build pipelines for multi-camera spatial fusion, occupancy analysis, or activity-zone mapping
- Evaluate on real deployment data from airports, buildings, and public spaces

Track B: VLM Optimization & Deployment

- Fine-tune and optimize vision-language models (e.g., Florence, Qwen-VL, InternVL) for Certis-specific tasks: anomaly description, scene captioning, visual Q&A for security operations
- Develop distillation, quantization, or pruning strategies for edge/real-time deployment
- Build evaluation benchmarks for VLM accuracy, latency, and robustness in operational conditions

Track C: Detection & Segmentation Enhancement

- Improve detection/segmentation models (YOLO, SAM, etc.) for domain-specific scenarios: crowded spaces, low-light, occlusion
- Build active learning or auto-labeling pipelines to accelerate data annotation
- Develop evaluation frameworks with operational metrics (false alarm rate, detection latency, cross-camera consistency)

Track D: Spatial-Temporal Behavior Analysis for SOP Compliance

- Develop models for understanding human activity patterns, movement trajectories, and behavioral sequences in operational environments
- Build spatial-temporal reasoning pipelines that map observed behaviors against defined Standard Operating Procedures (SOPs) — detecting deviations, missed steps, or non-compliance
- Integrate with video analytics systems to enable real-time or near-real-time compliance monitoring
- Design evaluation metrics grounded in operational requirements: detection accuracy, false alarm rate, temporal precision, and explainability of flagged events

Expected deliverables

- A working vision component or optimization pipeline integrated into the team's workflow: e.g., a spatial reasoning module, an optimized VLM, an improved detection pipeline, or a behavior analysis system
- Technical report: problem definition, methodology, experimental results, and deployment recommendations

Requirements

- Strong Python skills; experience with PyTorch
- Solid computer vision fundamentals (detection, segmentation, feature extraction)
- Familiarity with at least one relevant model family: YOLO variants, SAM, or VLMs
- Bonus: experience with model optimization (TensorRT, ONNX, quantization), 3D vision, multi-camera systems, or activity recognition

What we value

- Ability to bridge model research and deployment reality (latency, accuracy trade-offs, edge constraints)
- Strong experimental rigor: clear metrics, fair baselines, reproducible results
- Curiosity about real-world visual challenges (lighting, occlusion, camera variance, crowd density)

How to apply

Please send:

- Resume + GitHub/portfolio (if available)
- 1–2 paragraphs: your preferred track(s) and a relevant project or paper

岗位六：计算机视觉（空间智能与 VLM 优化）

岗位概述

我们正在推进 Certis AI 产品背后的视觉智能能力——包括 SeVU（智能视频分析）、Mozart、MIRA、VOXIS 和 IncidentIQ。本实习岗位聚焦于空间理解、视觉语言模型（VLM）优化、时空行为分析以及公共空间场景智能，推动视觉系统的感知、理解与推理能力边界。

方向选择（可选 1–2 个方向）

方向 A：空间智能与场景理解

- 开发或改进空间推理能力：深度估计、3D 场景理解、相机画面中的空间关系提取
- 构建多相机空间融合、占用分析、活动区域映射管道
- 基于机场、建筑、公共空间的真实部署数据进行评测

方向 B：VLM 优化与部署

- 针对 Certis 场景微调与优化视觉语言模型（如 Florence、Qwen-VL、InternVL）：异常描述、场景标注、安防运营视觉问答
- 开发蒸馏、量化、剪枝策略，支持边缘/实时部署
- 构建 VLM 准确率、延迟、运行环境鲁棒性的评测基准

方向 C：检测与分割增强

- 改进面向特定场景的检测/分割模型（YOLO、SAM 等）：人流密集、低光照、遮挡场景
- 构建主动学习或自动标注管道，加速数据标注效率
- 开发以运营指标为导向的评测框架（误报率、检测延迟、跨相机一致性）

方向 D：时空行为分析与 SOP 合规检查

- 开发人体活动模式、运动轨迹、行为序列理解模型
- 构建时空推理管道，将观察到的行为与定义的标准操作流程（SOP）对比——检测偏差、缺失步骤或不合规行为
- 与视频分析系统集成，实现实时或准实时合规监控
- 设计以运营需求为导向的评测指标：检测准确率、误报率、时间精度、标记事件的可解释性

预期产出

- 可集成到团队工作流的视觉组件或优化管道：例如空间推理模块、优化后的 VLM、改进的检测管道、或行为分析系统
- 技术报告：问题定义、方法论、实验结果与部署建议

基本要求

- 扎实的 Python 编程能力；具备 PyTorch 使用经验
- 良好的计算机视觉基础（检测、分割、特征提取）
- 熟悉以下至少一个模型系列：YOLO 变体、SAM、或 VLM
- 加分项：模型优化经验（TensorRT、ONNX、量化）、3D 视觉、多相机系统、或行为识别

我们看重的能力

- 能够连接模型研究与部署现实（延迟、精度权衡、边缘约束）
- 实验严谨：指标清晰、基线公平、结果可复现
- 对真实视觉挑战的好奇心（光照、遮挡、相机差异、人流密度）

申请方式

请提供：

- 简历 + GitHub/作品集（如有）
- 1–2 段说明：你最想做的方向（可多选）以及一个相关项目或论文